

Rapilin

Insulin Aspart 100 U/mL Solution for Injection

PRODUCT NAME

Rapilin Insulin Aspart 100 U/mL Solution for Injection

COMPOSITION

Each mL contains:

Insulin Aspart* (rDNA origin) 100 U (equivalent to 3.5 mg).

1 cartridge contains 3 mL equivalent to 300 U.

*Insulin Aspart is produced by recombinant DNA technology in *Escherichia coli*.

Contains M-cresol 1.72 mg and Phenol 1.50 mg as preservative.

Excipient: M-cresol, Glycerin, Phenol, Zinc chloride, Sodium phosphate dibasic anhydrous, Sodium chloride, Water for injection, Hydrochloric acid, Sodium hydroxide.

PRODUCT DESCRIPTION

The solution is clear, colourless and aqueous solution for injection in cartridge.

Rapilin is a biosimilar medicinal product of NovoRapid.

PHARMACODYNAMICS

Pharmacotherapeutic group: Drugs used in diabetes. Insulins and analogues for injection, fast acting.

ATC code: A10AB05.

Unless otherwise specified, data presented in this package insert for Efficacy and Safety were referred from package insert of NovoRapid.

Mechanism of action

Insulin aspart produces a more rapid onset of action compared to soluble human insulin, together with a lower glucose concentration, as assessed within the first four hours after a meal. Insulin aspart has a shorter duration of action compared to soluble human insulin after subcutaneous injection.

When insulin aspart is injected subcutaneously, the onset of action will occur within 10 to 20 minutes of injection. The maximum effect is exerted between 1 and 3 hours after injection. The duration of action is 3 to 5 hours.

Insulin aspart is equipotent to soluble human insulin on a molar basis.

Adults: clinical trials in patients with type 1 diabetes have demonstrated a lower postprandial blood glucose with insulin aspart compared to soluble human insulin. In two long-term open label trials in patients with type 1 diabetes comprising 1,070 and 884 patients, respectively, insulin aspart reduced glycated haemoglobin by 0.12 percentage points and by 0.15 percentage points compared to soluble human insulin; a difference of limited clinical significance.

Clinical trials in patients with type 1 diabetes have demonstrated a reduced risk of nocturnal hypoglycaemia with insulin aspart compared to soluble human insulin. The risk of daytime hypoglycaemia was not significantly increased.

Elderly: in a PK/PD trial the relative differences in the PD properties between insulin aspart and soluble human insulin in the elderly patients with type 2 diabetes were similar to those seen in healthy subjects and younger patients with diabetes.

Children and adolescents: when given to children, insulin aspart showed similar long-term glucose control compared to soluble human insulin.

In clinical trials for children and adolescents aged 2 to 17, the pharmacodynamic profile of insulin aspart in children was similar to that seen in adults.

The efficacy and safety of insulin aspart given as bolus insulin in combination with either insulin detemir or insulin degludec as basal insulin have been studied for up to 12 months in two randomised controlled clinical trials in adolescents and children aged 1 to less than 18 years (n=712). The trials included 167 children aged 1–5 years, 260 aged 6–11 and 285 aged 12–17. The observed improvements in HbA1c and the safety profiles were comparable between all age groups.

Pregnancy: a clinical trial comparing safety and efficacy of insulin aspart vs. soluble human insulin in the treatment of pregnant women with type 1 diabetes (322 exposed pregnancies) did not indicate any adverse effect of insulin aspart on pregnancy or on the health of the foetus/newborn.

In addition, the data from a clinical trial including 27 women with gestational diabetes randomised to treatment with insulin aspart vs. soluble human insulin showed similar safety profiles between treatments as well as a significant improvement in postprandial glucose control in the insulin aspart treated group.

Study GL-ASP-1008: A Phase I study in healthy subjects to compare PK/PD between Rapilin, NovoLog and NovoRapid has shown equivalence between 3 products was comparable. Based on the primary PK endpoints, $AUC_{ins,0-12h}$ and $C_{ins,max}$, PK equivalence of Rapilin versus NovoRapid and NovoLog was demonstrated since the 90% CIs of the respective geometric LS-mean treatment ratio lies within the limits of 80.00% and 125.00%. Results of secondary PK AUC endpoints further supported PK equivalence of Rapilin to NovoRapid and to NovoLog. Based on the primary PD endpoints $AUC_{GIR,0-12h}$ and GIR_{max} , PD equivalence of Rapilin versus NovoRapid, Rapilin versus NovoLog and of NovoLog versus NovoRapid was achieved since the 95% or 90% CIs of the respective LS-mean treatment ratio of untransformed data lie within the limits of 80.00% to 120.00%. Results of the primary PD analysis were robust and were supported by secondary and sensitivity PD analyses using log-transformed data and/or excluding profiles due to pre-defined criteria. The results of secondary PD AUC endpoints further supported PD equivalence of Rapilin to NovoRapid and to NovoLog. The overall results of this study show that Rapilin is equivalent to NovoRapid and to NovoLog with regard to the primary and secondary PK and PD endpoints.

Study 2011L01347: A Phase III study in T2DM patients to demonstrated equivalence in safety, immunogenicity and efficacy profile between Rapilin and NovoRapid. A 24-week-long clinical study indicates that Rapilin developed by Gan & Lee Pharmaceutical Co., Ltd. and NovoRapid treatment methods can enable patients with type 2 diabetes to achieve good blood glucose control. Using HbA1c, fasting blood glucose, and 2-hr post-meal venous blood glucose levels as determination criteria, treatment with Rapilin is non-inferior to NovoRapid in therapeutic effect for diabetic patients. There were no differences between the two groups in terms of hypoglycemic event incidence rates, adverse event incidence rates, laboratory test (blood routine, urine routine, blood biochemistry) abnormalities, and ECG abnormalities. The study results demonstrate that 3x/day use of Rapilin to treat diabetes is safe and effective and features good tolerance.

PHARMACOKINETICS

In insulin aspart substitution of amino acid proline with aspartic acid at position B28 reduces the tendency to form hexamers as observed with soluble human insulin. Insulin aspart is therefore more rapidly absorbed from the subcutaneous layer compared to soluble human insulin.

The time to maximum concentration is, on average, half of that for soluble human insulin. A mean maximum plasma concentration of 492 pmol/l was reached 40 minutes after a subcutaneous dose of 0.15 unit/kg bodyweight in type 1 diabetic patients. The insulin concentrations returned to baseline about 4 to 6 hours after dose. The absorption rate was somewhat slower in type 2 diabetic patients, resulting in a lower C_{max} (352 ± 240 pmol/l) and later t_{max} (60 minutes). The intra-individual variability in time to maximum concentration is significantly less for Insulin Aspart than for soluble human insulin, whereas the intraindividual variability in C_{max} for Insulin Aspart is larger.

Children and adolescents: the pharmacokinetic and pharmacodynamic properties of insulin aspart were investigated in children and adolescents with type 1 diabetes. Insulin aspart was rapidly absorbed in both age groups, with similar $t_{\max}$ as in adults. However, $C_{\max}$ differed between the age groups, stressing the importance of the individual titration of insulin aspart.

Elderly: the relative differences in pharmacokinetic properties between insulin aspart and soluble human insulin in elderly patients with type 2 diabetes were similar to those observed in healthy subjects and in younger patients with diabetes. A decreased absorption rate was observed in elderly patients, resulting in a later $t_{\max}$ (82 minutes), whereas $C_{\max}$ was similar to that observed in younger patients with type 2 diabetes and slightly lower than in patients with type 1 diabetes.

Hepatic impairment: in patients with hepatic impairment, $t_{\max}$ was delayed to about 85 min. (50 min. in subjects with normal hepatic function) while AUC, $C_{\max}$ and CL/F were similar.

Renal impairment: a single dose pharmacokinetic study of insulin aspart in 18 subjects with normal to severely impaired renal function was performed. No apparent effect of creatinine clearance values on AUC, $C_{\max}$, CL/F and $t_{\max}$ of insulin aspart was found. Data were limited in patients with moderate and severe renal impairment. Patients with renal failure necessitating dialysis treatment were not investigated.

INDICATION

Rapilin is indicated for treatment of diabetes mellitus in adults, adolescents and children aged 1 year and above.

DOSAGE AND ADMINISTRATION

Posology

Insulin aspart is a rapid-acting insulin analogue. Insulin aspart dosage is individual and determined in accordance with the needs of the patient. It should normally be used in combination with intermediate-acting or long-acting insulin given at least one a day. Blood glucose monitoring and insulin dose adjustment are recommended to achieve optimal glycaemic control.

The individual insulin requirement in adults and children is usually between 0.5 and 1.0 U/kg/day. In a basal-bolus treatment regimen, 50–70% of this requirement may be provided by insulin aspart and the remainder by intermediate-acting or long-acting insulin. Adjustment of dosage may be necessary if patients undertake increased physical activity, change their usual diet or during concomitant illness.

Insulin aspart has a faster onset and a shorter duration of action than soluble human insulin.

Due to the faster onset of action, insulin aspart should generally be given immediately before a meal. When necessary insulin aspart can be given soon after a meal.

Due to the shorter duration, insulin aspart has a lower risk of causing nocturnal hypoglycaemic episodes.

Special populations

As with all insulin products, in elderly patients and patients with renal or hepatic impairment, glucose monitoring should be intensified and the insulin aspart dosage adjusted on an individual basis.

Paediatric population

Insulin aspart can be used in children and adolescents aged 1 year and above in preference to soluble human insulin when a rapid onset of action might be beneficial, for example, in the timing of the injections in relation to meals. The safety and efficacy of insulin aspart in children below 1 year of age have not been established. No data are available.

Transfer from other insulin products

When transferring from other insulin products, adjustment of the insulin aspart dose and the dose of the basal insulin may be necessary.

Method of Administration

Insulin aspart is administered subcutaneously by injection in the abdominal wall, the thigh, the upper arm, the deltoid region or the gluteal region. Injection sites should always be rotated within the same region in order to reduce the risk of lipodystrophy and cutaneous amyloidosis. As with all insulin products, subcutaneous injection in the abdominal wall ensures a faster absorption than other injection sites.

The duration of action will vary according to the dose, injection site, blood flow, temperature and level of physical activity. However, the faster onset of action compared to soluble human insulin is maintained regardless of injection site.

The Rapilin cartridge is designed to be used with reusable pen i.e., GanleePen. Detailed instruction accompanying the delivery system must be followed.

Instruction for Use

Please bring Rapilin to room temperature before use and follow the injection steps described below:

1. Check the insulin to make sure it is clear and colourless. Do not use Rapilin if it is coloured or cloudy, or if there are particles in the solution.
2. Please refer to the relevant reusable pen package insert, when using Rapilin in conjunction with a reusable pen.
3. Rapilin is injected under the skin (subcutaneously) of the upper arm, thigh, or stomach area (abdomen). Wipe the skin with an alcohol swab to clean the injection site. Let the injection site dry before injection.
4. Pinch the skin of the injection site with fingers and puncture the needle. After the pusher is pushed to the end, keep the needle under the skin for a few seconds to ensure that the correct dose is injected, then pull out the needle, and gently press the injection site with the sterile cotton ball for a few seconds, but do not rub the injection site to avoid damage to the subcutaneous tissue or cause the exudation of Rapilin.

CONTRAINDICATIONS

Hypersensitivity to the active substance or any of the excipients.

WARNING AND PRECAUTIONS

Before travelling between different time zones, the patients should seek the doctor's advice since this may mean that the patient has to take the insulin and meals at different times.

Hyperglycaemia

Inadequate dosing or discontinuation of treatment, especially in type 1 diabetes, may lead to hyperglycaemia and diabetic ketoacidosis.

Hypoglycaemia

Omission of a meal or unplanned, strenuous physical exercise may lead to hypoglycaemia.

Especially in children, care should be taken to match insulin doses (especially in basal-bolus regimens) with food intake, physical activities and current blood glucose level in order to minimise the risk of hypoglycaemia.

Hypoglycaemia may occur if the insulin dose is too high in relation to the insulin requirement.

Patients whose blood glucose control is greatly improved, e.g. by intensified insulin therapy, may experience a change in their usual warning symptoms of hypoglycaemia, and should be advised accordingly. Usual warning symptoms

may disappear in patients with longstanding diabetes.

A consequence of the pharmacodynamics of rapid-acting insulin analogues is that if hypoglycaemia occurs, it may occur earlier after an injection when compared to soluble human insulin.

Since insulin aspart should be administered in immediate relation to a meal, the rapid onset of action should be considered in patients with concomitant diseases or medication where a delayed absorption of food might be expected.

Concomitant illness, especially infections and feverish conditions, usually increases the patient's insulin requirements. Concomitant diseases of the kidney, liver or affecting the adrenal, pituitary or thyroid gland can require changes in the insulin dose.

When patients are transferred between different types of insulin products, the early warning symptoms of hypoglycaemia may become less pronounced than those experienced with their previous insulin.

Transfer from other insulin medicinal products

Transferring a patient to another type or brand (e.g. strength or manufacturer) of insulin should be done under strict medical supervision and may require a change in dosage or number of daily injections from that used with their usual insulin products. If an adjustment is needed, it may occur with the first dose or during the first few weeks or months.

Injection site reactions

As with any insulin therapy, injection site reactions may occur and include pain, redness, hives, inflammation, bruising, swelling and itching. Continuous rotation of the injection site within a given area reduces the risk of developing these reactions. Reactions usually resolve in a few days to a few weeks. On rare occasions, injection site reactions may require discontinuation of insulin aspart.

Skin and subcutaneous tissue disorders

Patients must be instructed to perform continuous rotation of the injection site to reduce the risk of developing lipodystrophy and cutaneous amyloidosis. There is a potential risk of delayed insulin absorption and worsened glycaemic control following insulin injections at sites with these reactions. A sudden change in the injection site to an unaffected area has been reported to result in hypoglycaemia. Blood glucose monitoring is recommended after the change in the injection site from an affected to an unaffected area, and dose adjustment of antidiabetic medications may be considered.

Combination of thiazolidinediones and insulin medicinal products

Cases of congestive heart failure have been reported when thiazolidinediones were used in combination with insulin, especially in patients with risk factors for development of congestive heart failure. This should be kept in mind if treatment with the combination of thiazolidinediones and insulin medicinal products is considered. If the combination is used, patients should be observed for signs and symptoms of congestive heart failure, weight gain and oedema. Thiazolidinediones should be discontinued if any deterioration in cardiac symptoms occurs.

Avoidance of accidental mix-ups/medication errors

Patients must be instructed to always check the insulin label before each injection to avoid accidental mix-ups between insulin aspart and other insulin products.

Insulin antibodies

Insulin administration may cause insulin antibodies to form. In rare cases, the presence of such insulin antibodies may necessitate adjustment of the insulin dose in order to correct a tendency to hyperglycaemia or hypoglycaemia.

INTERACTIONS WITH OTHER MEDICAMENTS

A number of medicinal products are known to interact with the glucose metabolism.

Gan & Lee Pharmaceuticals

The following substances may reduce the patient's insulin requirements:

Oral antidiabetic products, monoamine oxidase inhibitors (MAOI), beta-blockers, angiotensin converting enzyme (ACE) inhibitors, salicylates, anabolic steroids and sulphonamides.

The following substances may increase the patient's insulin requirements:

Oral contraceptives, thiazides, glucocorticoids, thyroid hormones, sympathomimetics, growth hormone and danazol.

Beta-blockers may mask the symptoms of hypoglycaemia.

Octreotide/lanreotide may either increase or decrease the insulin requirement.

Alcohol may intensify or reduce the hypoglycaemic effect of insulin.

PREGNANCY AND LACTATION

Pregnancy

Insulin aspart can be used in pregnancy. Data from two randomized controlled clinical trials do not indicate any adverse effect of insulin aspart on pregnancy or on the health of the foetus/newborn when compared to soluble human insulin.

Intensified blood glucose control and monitoring of pregnant women with diabetes are recommended throughout pregnancy and when contemplating pregnancy. Insulin requirements usually fall in the first trimester and increase subsequently during the second and third trimesters. After delivery, insulin requirements normally return rapidly to pre-pregnancy values.

Lactation

There are no restrictions on treatment with insulin aspart during breast-feeding. Insulin treatment of the nursing mother presents no risk to the baby. However, the insulin aspart dosage may need to be adjusted.

ADVERSE REACTIONS

a. Summary of the safety profile

Adverse reactions observed in patients using insulin aspart are mainly due to the pharmacologic effect of insulin.

The most frequently reported adverse reaction during treatment is hypoglycaemia. The frequencies of hypoglycaemia vary with patient population, dose regimens and level of glycaemic control, please see section c below.

At the beginning of the insulin treatment, refraction anomalies, oedema and injection site reactions (pain, redness, hives, inflammation, bruising, swelling and itching at the injection site) may occur. These reactions are usually of transitory nature. Fast improvement in blood glucose control may be associated with acute painful neuropathy, which is usually reversible. Intensification of insulin therapy with abrupt improvement in glycaemic control may be associated with temporary worsening of diabetic retinopathy, while long-term improved glycaemic control decreases the risk of progression of diabetic retinopathy.

b. Tabulated list of adverse reactions

Adverse reactions listed below are based on clinical trial data and classified according to MedDRA System Organ Class. Frequency categories are defined according to the following convention: Very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$); not known (cannot be estimated from the available data).

Immune system disorders	Uncommon – Urticaria, rash, eruptions
	Very rare – Anaphylactic reactions*

Metabolism and nutrition disorders	Very common – Hypoglycaemia*
Nervous system disorders	Rare – Peripheral neuropathy (painful neuropathy)
Eye disorders	Uncommon – Refraction disorders
	Uncommon – Diabetic retinopathy
Skin and subcutaneous tissue disorders	Uncommon – Lipodystrophy*
	Not known – Cutaneous amyloidosis*†
General disorders and administration site conditions	Uncommon – Injection site reactions
	Uncommon – Oedema

*See below for descriptions of selected adverse reactions.

† ADR from post-marketing sources.

c. Description of selected adverse reactions

Anaphylactic reactions

The occurrence of generalised hypersensitivity reactions (including generalised skin rash, itching, sweating, gastrointestinal upset, angioneurotic oedema, difficulties in breathing, palpitation and reduction in blood pressure) is very rare but can potentially be life threatening.

Hypoglycaemia

The most frequently reported adverse reaction is hypoglycaemia. It may occur if the insulin dose is too high in relation to the insulin requirement. Severe hypoglycaemia may lead to unconsciousness and/or convulsions and may result in temporary or permanent impairment of brain function or even death. The symptoms of hypoglycaemia usually occur suddenly. They may include cold sweats, cool pale skin, fatigue, nervousness or tremor, anxiousness, unusual tiredness or weakness, confusion, difficulty in concentration, drowsiness, excessive hunger, vision changes, headache, nausea and palpitation.

In clinical trials, the frequency of hypoglycaemia varied with patient population, dose regimens and level of glycaemic control. During clinical trials the overall rates of hypoglycaemia did not differ between patients treated with insulin aspart compared to human insulin.

Skin and subcutaneous tissue disorders

Lipodystrophy (including lipohypertrophy, lipoatrophy) and cutaneous amyloidosis may occur at the injection site and delay local insulin absorption. Continuous rotation of the injection site within the given injection area may help to reduce or prevent these reactions.

SYMPTOMS AND TREATMENT OF OVERDOSE

A specific overdose for insulin cannot be defined, however, hypoglycaemia may develop over sequential stages if too high doses relative to the patient's requirement are administered:

- Mild hypoglycaemic episodes can be treated by oral administration of glucose or sugary products. It is therefore recommended that the diabetic patient always carries sugar-containing products.
- Severe hypoglycaemic episodes, where the patient has become unconscious, can be treated with glucagon (0.5 to 1 mg) given intramuscularly or subcutaneously by a trained person, or with glucose given intravenously by physicians or other healthcare staff if applicable. Glucose must be given intravenously, if the patient does not respond to glucagon within 10 to 15 minutes. Upon regaining consciousness, administration of oral carbohydrates is recommended for the patient in order to prevent a relapse.

EFFECT ON ABILITY TO DRIVE AND USE MACHINES

The patient's ability to concentrate and react may be impaired as a result of hypoglycaemia. This may constitute a risk in situations where these abilities are of special importance (e.g., driving a car or operating machinery).

Patients should be advised to take precautions to avoid hypoglycaemia while driving. This is particularly important in those who have reduced or absent awareness of the warning signs of hypoglycaemia or have frequent episodes of hypoglycaemia. The advisability of driving should be considered in these circumstances.

PRECLINICAL SAFETY DATA

Non clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity and toxicity to reproduction and development.

In *in vitro* tests, including binding to insulin and IGF-1 receptor sites and effects on cell growth, insulin aspart behaved in a manner that closely resembled human insulin. Studies also demonstrate that the dissociation of binding to the insulin receptor of insulin aspart is equivalent to human insulin.

INCOMPATIBILITIES

Substances added to Rapilin may cause degradation of insulin aspart. This product must not be diluted or mixed with other products.

STORAGE CONDITIONS

Unopened cartridges

Store in a refrigerator (2°C -8°C). Do not freeze.

Keep away from the cooling element. Keep the cartridge in the outer carton in order to protect from light.

In-use cartridges

Do not refrigerate. Store below 30°C. Use within 4 weeks.

Do not freeze. Do not expose from excessive heat or direct light.

SHELF LIFE

Unopened cartridges

30 months.

In-use cartridges

28 days.

DOSAGE FORM AND PACKAGING AVAILABLE

3 mL solution in a cartridge (Type I colourless glass) with a bromobutyl plunger and a cap comprising a rubber liner (rubber) and an aluminum seal.

1 cartridge/box.

MANUFACTURER

Gan & Lee Pharmaceuticals

No. 8 Nanfeng West First Road,

Gan & Lee Pharmaceuticals

Tongzhou District, Huoxian Town,
Beijing, 101109, China.

PRODUCT REGISTRATION HOLDER

GoodScience Sdn Bhd

No. 7, Jalan KPK 4/3,
Kawasan Perindustrian Kundang, Kundang Jaya,
48020 Rawang, Selangor, Malaysia.

DATE OF REVISION

8 January 2026